Mid Semester Report

Design and Development of Flexible Thermoelectric Devices

BY

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Under the supervision of

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ABSTRACT

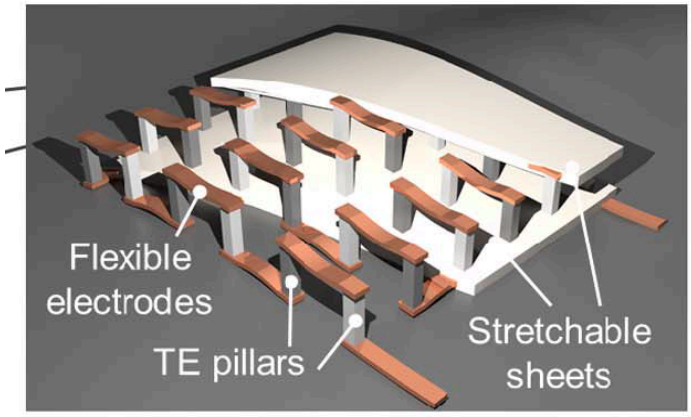
Flexible thermoelectric devices (TEDs) have emerged as promising candidates for energy harvesting from human body heat to power wearable electronics. The development of these devices hinges on balancing flexibility, efficiency, and comfort while maintaining adequate electrical and thermal performance. This report reviews existing approaches in the design and fabrication of flexible TEDs, focusing on materials, structural innovations, and their applications. Furthermore, a simulation model will be presented to analyze the physical properties and efficiency of these devices.To validate the simulations, hand calculations were performed to cross-check key parameters such as thermal conductivity and the Seebeck coefficient. The results demonstrate the potential of flexible thermoelectric devices for wearable cooling applications, with the substrate offering both adaptability and effective thermal management.

1. Introduction

A thermoelectric device (TED) generates voltage when a temperature difference exists between its two sides, a phenomenon known as the thermoelectric effect. This effect allows for the direct conversion of temperature differences into electricity (known as the **Seebeck Effect**) or vice versa. TEDs are particularly appealing for personalized cooling applications because they are solid-state devices with compact designs, and their cooling power can be easily adjusted.

In this project, we aim to design a flexible and wearable TED capable of providing long-term cooling with a substantial cooling effect. Our innovative design features rigid, high-performance (high-ZT) thermoelectric pillars sandwiched between two stretchable elastomer sheets. This TED will also be integrated into a wearable garment, enabling efficient and sustained cooling and heating effects directly on the skin.

To make this possible, we're using an innovative fabrication method that includes a double layer of elastomer with an insulating air gap in between. The thermoelectric pillars are carefully designed with the perfect shape and spacing to maximize flexibility and minimize heat loss. This clever approach ensures the device remains flexible, durable, and highly efficient, paving the way for a whole new level of wearable cooling technology.



Wearable thermoelectric device (TED)

2. Development of Flexible TED Design

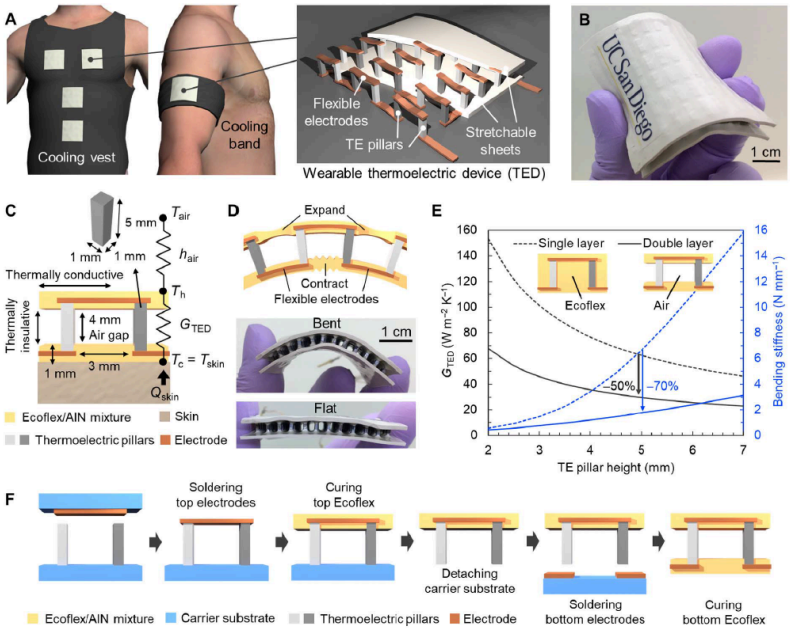
A new flexible and wearable TED is designed and demonstrated to show long-term active cooling greater than 8 hours with a very significant cooling effect of more than 10°C without the requirement of substantial heat sinks. The main advancements developed in the design are:

• Sandwich Structure: The flexibility is saved while incorporating air gap with the help of an inorganic TE pillar between two stretchable elastomer sheets called Ecoflex in order to achieve least thermal leakage.

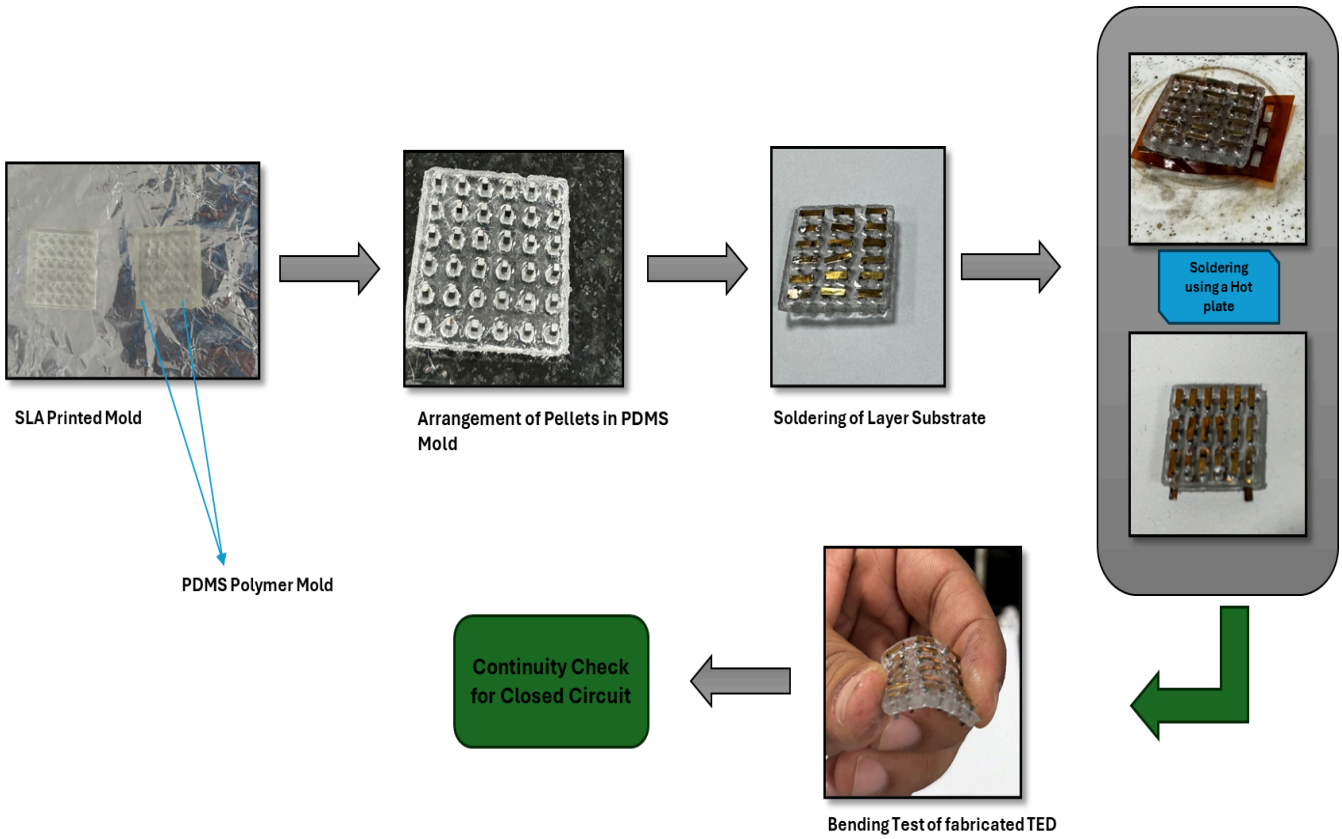
• Optimized TE Pillar Structure: The TE pillars have an aspect ratio, height-to-width, of 5:1 and a spatial density of 6.25%. This limits the major heat conduction along the cold and hot sides. The cross-sectional dimension of the pillars is kept at 1 mm by 1 mm, whereas the height is 5 mm with a separation distance of 3 mm.

We change the height of the legs and the pitch (the distance between 2 legs) later while doing simulation so as to get different results.

• Much thicker air gap between flexible sheets: The large air gap of 4 mm further reduces the thermal conductance. Top and bottom parts, 0.5 mm each, of the TE pillars are encapsulated in the Ecoflex sheets to ensure structural strength with flexibility.



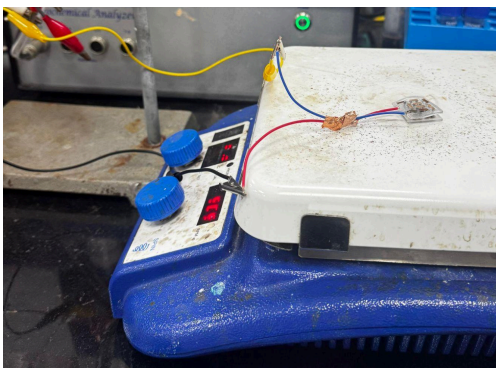
3. Fabrication of TED



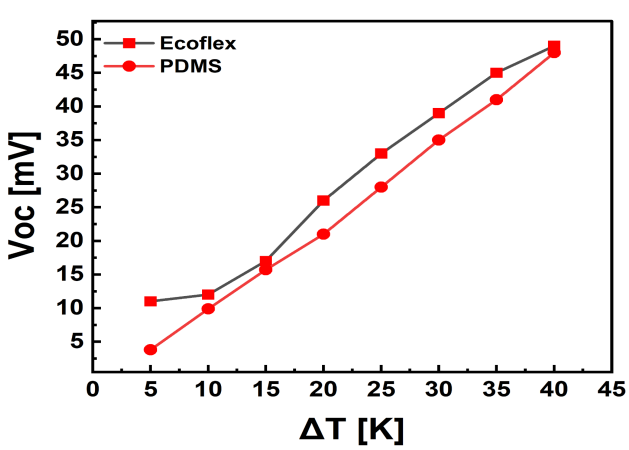
4. Results

a) Calculating the Open Circuit Voltage

For this experimental setup, we have put the fabricated TED device on a hot plate such that one side comes in contact with the hot plate. This creates a temperature difference between both sides and according to the Seebeck effect, an open circuit voltage is generated. This voltage was measured at different temperature differences and a graph was plotted for the same.



The setup was done for both the Ecoflex based TED and the PDMS based TED and the results were compared which showed that V_{OC} for Ecoflex was more than PDMS.



b) Calculating the Inner Resistance

The internal resistance (R) is a crucial factor in wearable thermoelectric generators (TEGs), significantly impacting their voltage and power output for heat harvesting applications. To determine R, one can utilize the measured open-circuit voltage (V_{OC}) and loading voltage (V_L) values. The formula for calculating this inner resistance is

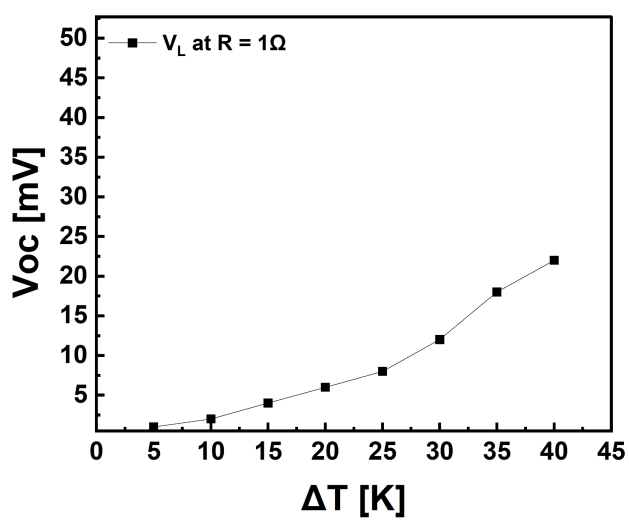
$$R = \frac{(V_{OC} - V_L) \times R_L}{V_L}$$

where R_L is the loading resistance.

The output power (P) that have been widely utilized to study the performance of TEG devices in heat harvesting can be calculated as

$$P = \frac{V_{OC}^2}{(R + R_L)^2} \times R_L$$

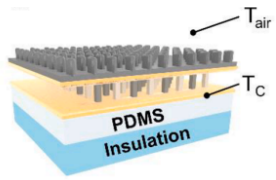
For our experimental setup we have used R_L as 1Ω.



Final calculations from the values obtained from the above graph showed that **inner resistance** was equal to **2.47 Ω** and the **output power** was equal to **30.46μW/cm^2.**

(C) Calculating Temperature Change versus Input Current

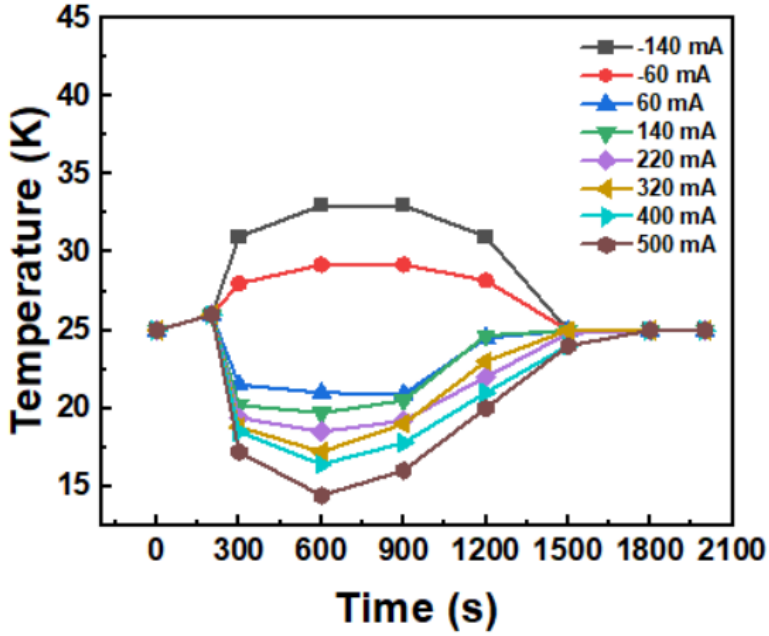
In this experimental configuration, the device was subjected to a range of electrical currents spanning from -140 mA to 500 mA, while the surrounding temperature was maintained at 22.5°C. To observe and record the thermal behavior, a thermal imaging camera was employed to capture images and monitor the temperature variations on the upper surface of the PDMS material throughout the experiment.



When the device was activated, the Peltier effect caused heat to move from the PDMS material to the surrounding air. This heat transfer led to a decrease in the PDMS temperature until it reached a stable state. After switching off the device, some of the previously transferred heat quickly returned, causing the temperature to slowly rise back towards room temperature.
In a similar manner, reversing the current direction altered the heat flow pattern. This reversal resulted in the PDMS experiencing a temperature rise until it stabilized. Once the device was deactivated, the temperature gradually declined, returning to room temperature over time.

The temperature change versus inputting current values were recorded in a table and then was plotted in a graph.

Time (Seconds)	Temperature (Celcius) at different Current					
	-140mA	-60mA	60mA	140mA	320mA	500mA
0	25	25	25	25	25	25
300	31	28	21.5	20.2	18.8	17.2
600	33	29.2	21	19.7	17.2	14.4
900	33	29.2	20.9	20.5	19	16
1200	31	28.2	24.5	24.6	23	20
1500	25	25	25	24	24.8	25
1800	25	25	25	25	25	25



From the above results, we checked that the experimental data fit well with the simulated data, proving that our modeling is reliable.

5. Conclusion

The experimental findings confirm the successful design and performance evaluation of a flexible thermoelectric device (TED). TheTED exhibited a high open-circuit voltage indicating improved thermoelectric efficiency. The inner resistance was calculated as 2.47 Ω, and the device achieved an output power of 30.46 μW/cm², demonstrating its capability for efficient heat harvesting. Additionally, the temperature response to varying input currents validated the TED's ability to regulate heat effectively through the Peltier effect. The cooling and heating behavior of the PDMS layer showed a stable temperature change, aligning well with simulated data, reinforcing the accuracy of the theoretical model. These results highlight the feasibility of using flexible TEDs for wearable applications, ensuring long-term thermal regulation.

The study also provides a strong foundation for further optimization to enhance power generation and thermal performance for practical energy-efficient cooling and heating solutions in wearable electronics.

6. References

- Q. Wu, J. Hu, A novel design for a wearable thermoelectric generator based on 3D fabric structure. *Smart Mater. Struct.* 26, 045037 (2017).
- J. He, T. M. Tritt, Advances in thermoelectric materials research: Looking back and moving forward. *Science* 357, eaak9997 (2017).
- J.-H. Bahk, H. Fang, K. Yazawa, A. Shakouri, Flexible thermoelectric materials and device optimization for wearable energy harvesting. *J. Mater. Chem. C* 3, 10362–10374 (2015).
- Yancheng Wang, Yaoguang Shib, Deqing Meia, Zichen Chen, Wearable thermoelectric generator to harvest body heat for powering a miniaturized accelerometer (2018).
- C. S. Kim, G. S. Lee, H. Choi, Y. J. Kim, H. M. Yang, S. H. Lim, S.-G. Lee, B. J. Cho, Structural design of a flexible thermoelectric power generator for wearable applications. *Appl. Energy* 214, 131–138 (2018).
- Pengcheng Zhu, Xuepeng Luo, Xinran Lin, Zhichong Qiu, Rongrui Chen, Xiangcheng Wang, Yaling Wang, Yuan Deng , Yanchao Mao, A self-healable, recyclable, and flexible thermoelectric device for wearable energy harvesting and personal thermal management